

Figure S1: The proportion of non-human samples across 100 studies encompassing 144,862 samples in the D1 dataset

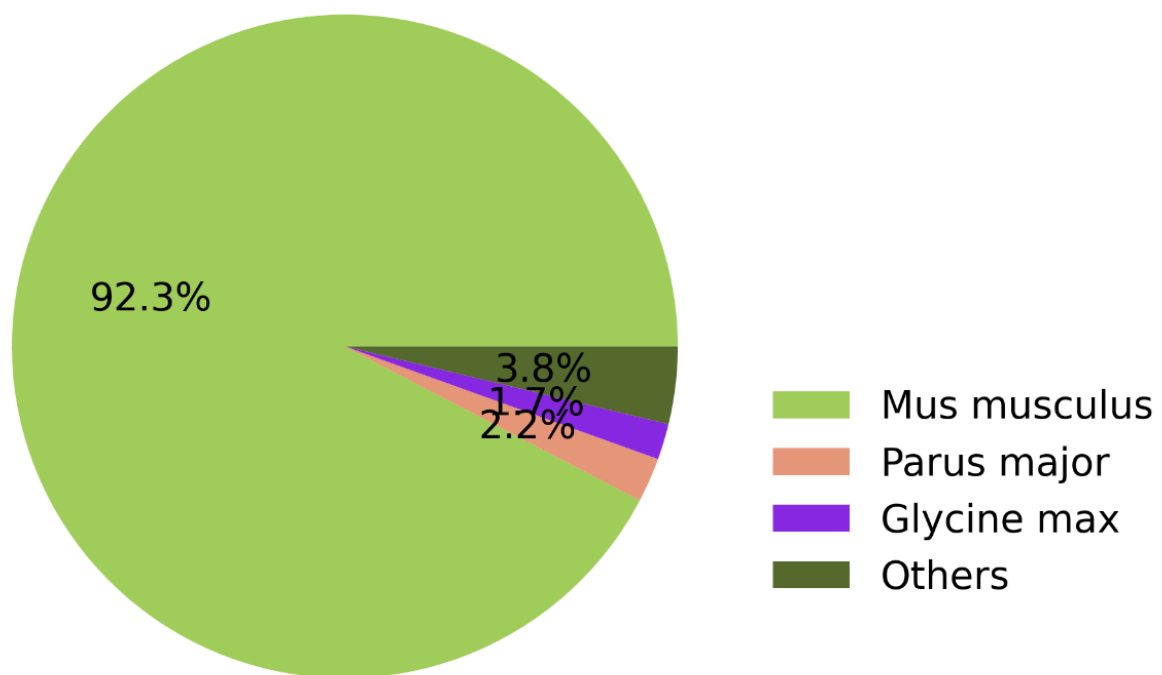


Figure S2: The metadata availability reported in the D1-dataset from the textual content of publications and public repositories. (Dotted line: the mean metadata availability among the samples in the D1-dataset.)

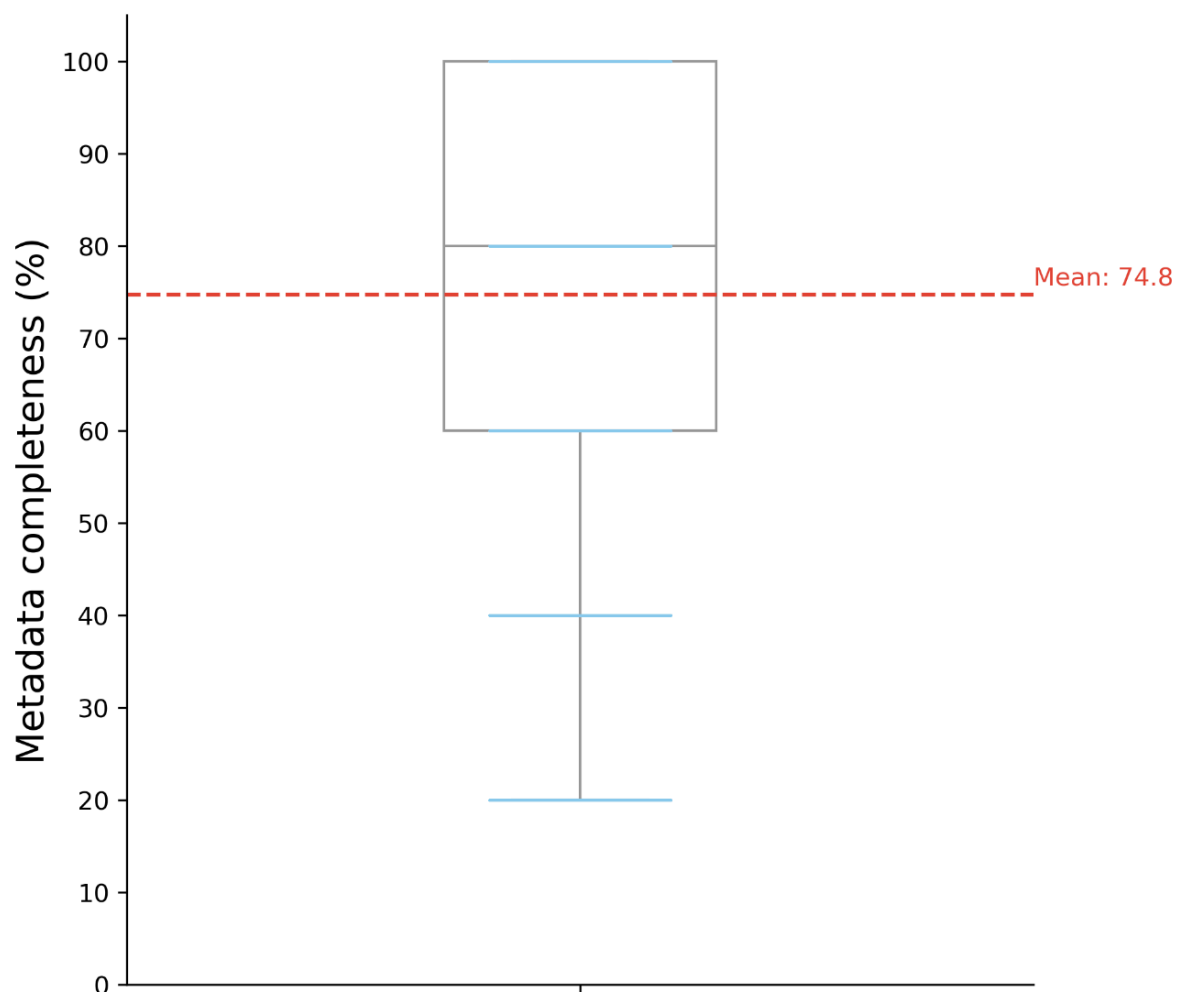


Figure S3: The cumulative metadata availability of the six essential metadata in the D1-dataset over the years.

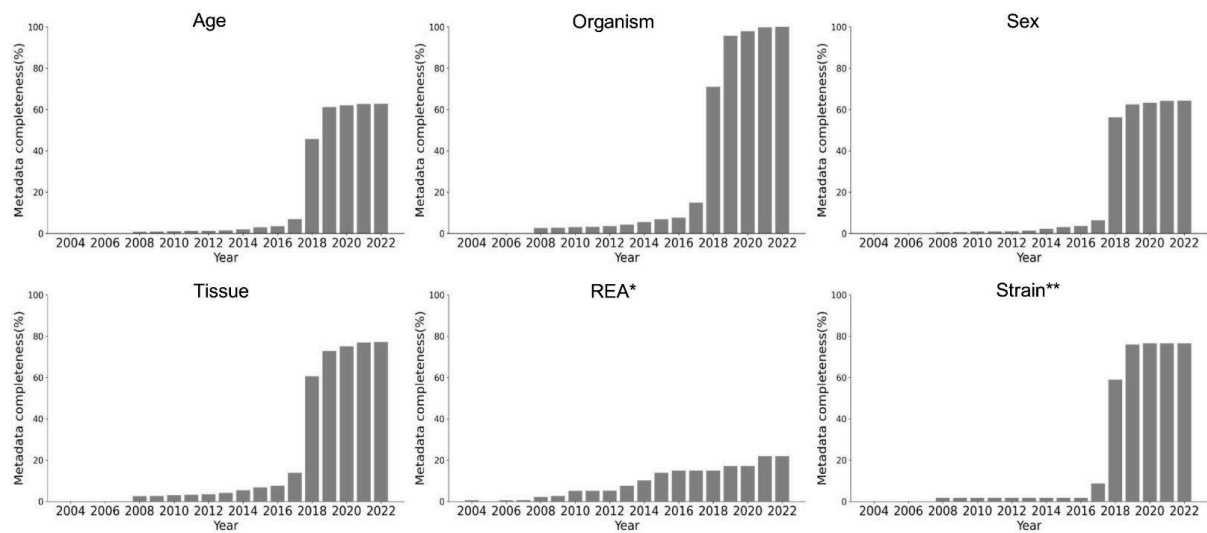


Figure S4: The human and non-human metadata availability in the textual content of publications (Original publication) and public repositories for the D1-dataset.

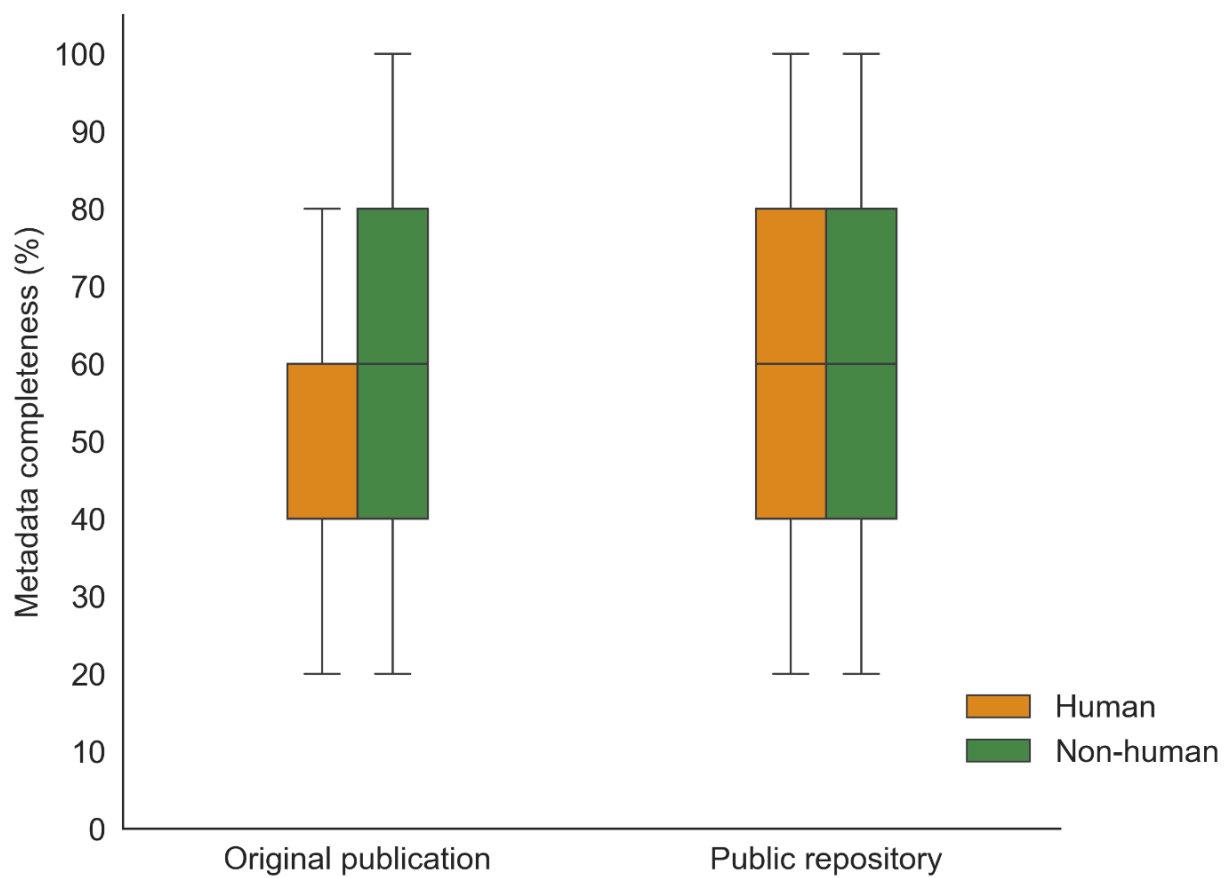


Figure S5: The metadata availability between the textual content of publications (Original Publication) and public repositories among the top five non-human organisms represented in the D1 dataset.

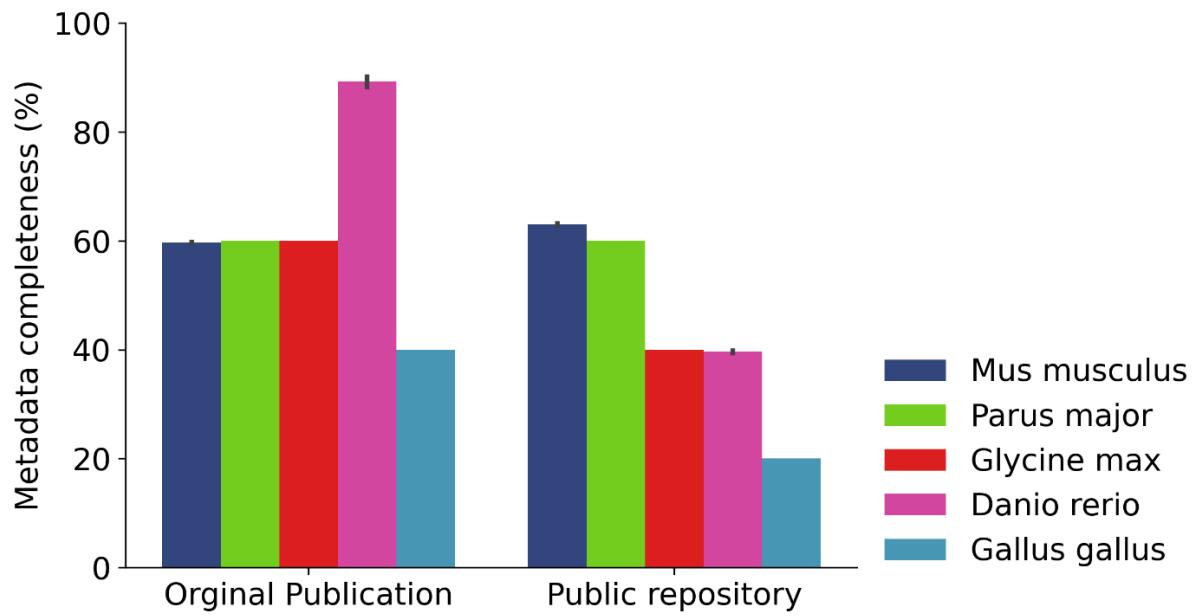


Figure S6: The comparison of the availability of sample-level and experiment-level metadata between human samples and non-human samples in the D1-dataset.

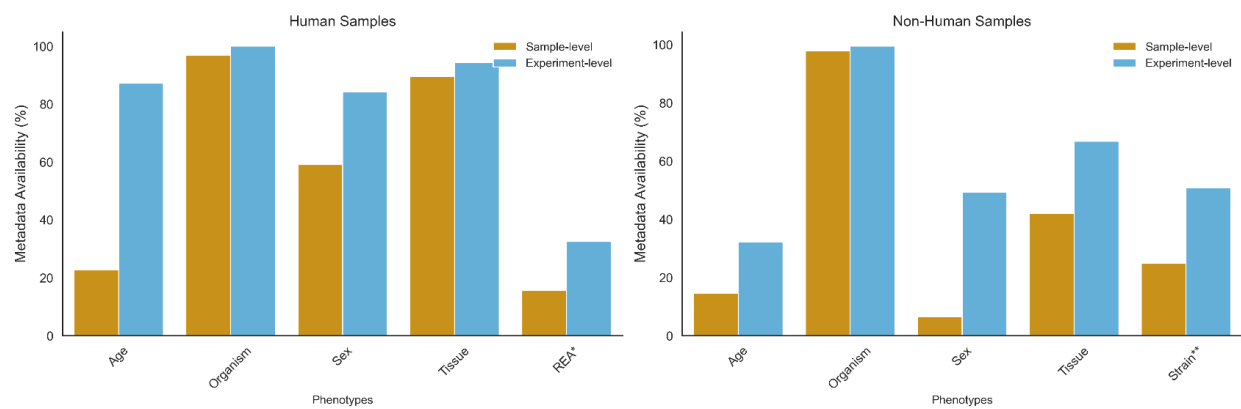


Figure S7: Right: The overall composition of sample-level and experiment-level metadata availability in the textual content of publications (OP) and public repositories (PR) across the human samples in the D1-dataset; Left: The overall composition of sample-level and experiment-level metadata availability in the textual content of publications and public repositories across the non-human samples in the D1-dataset.

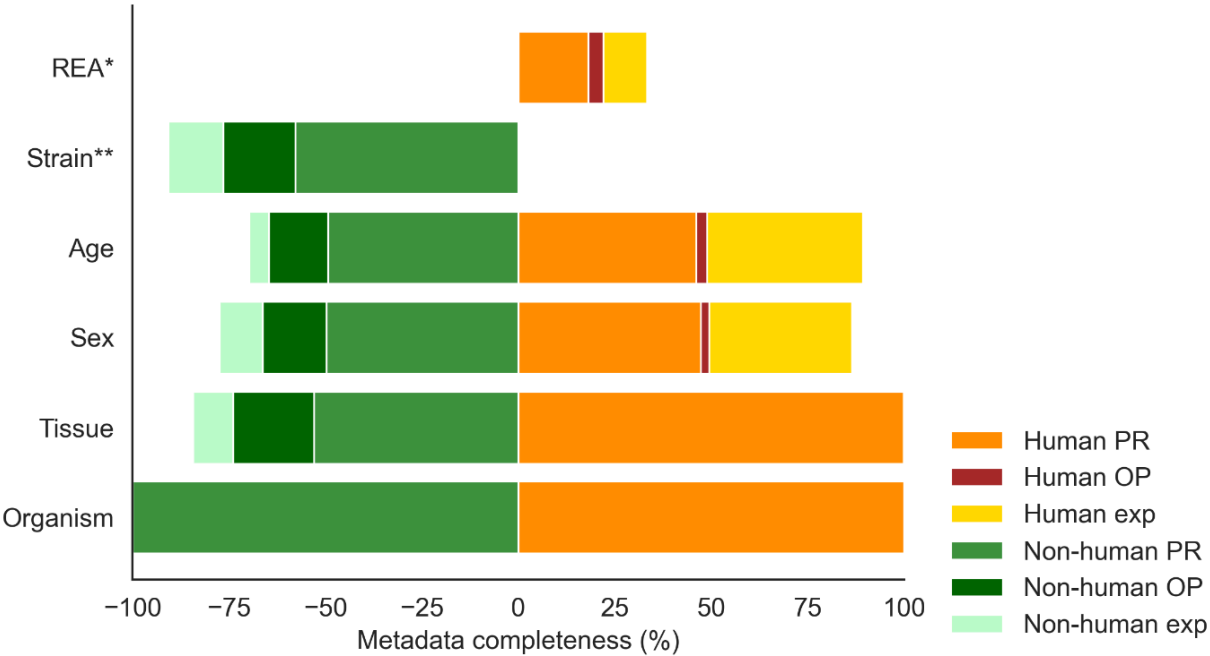


Figure S8: The comparison of the human and non-human metadata availability in the D2-dataset.

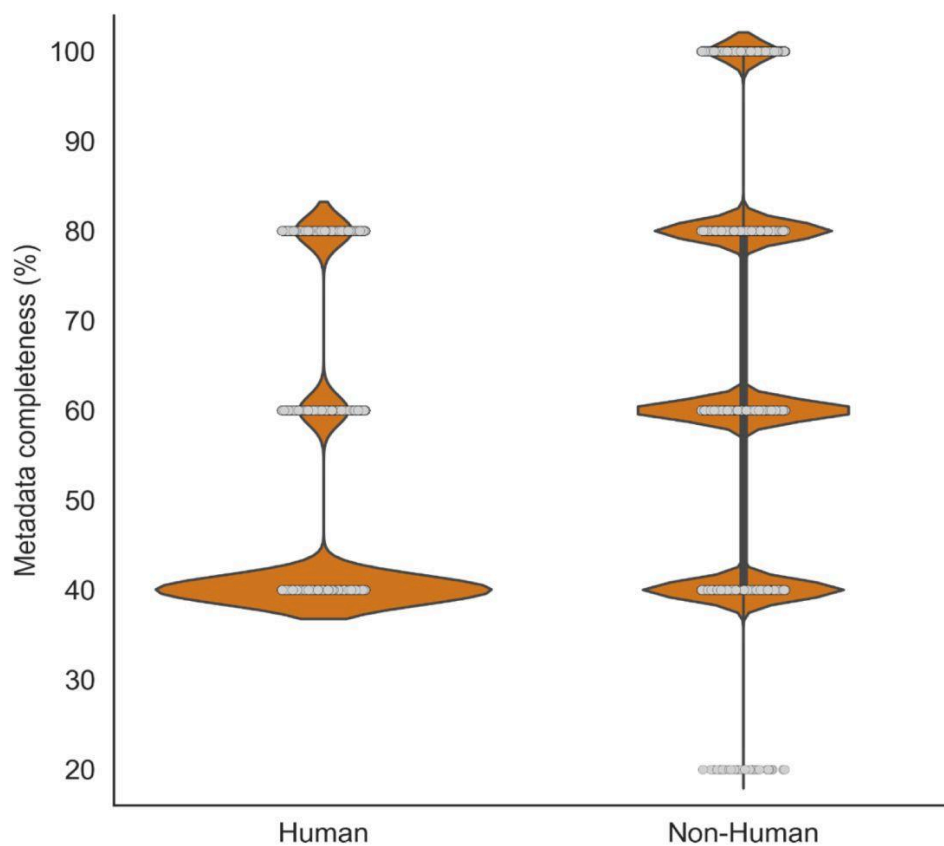


Figure S9: The cumulative metadata availability of the D2-dataset over the years.

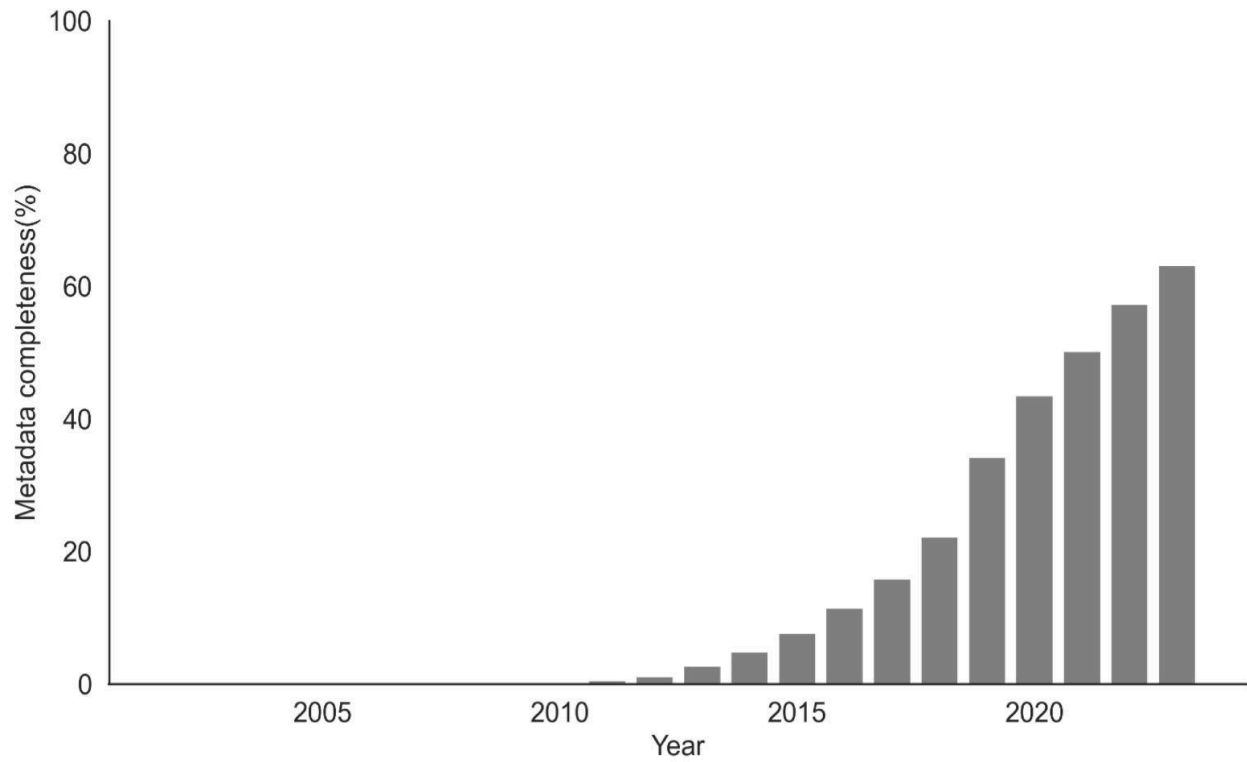


Figure S10: The separate metadata availability of human and non-human samples in the D2-dataset.

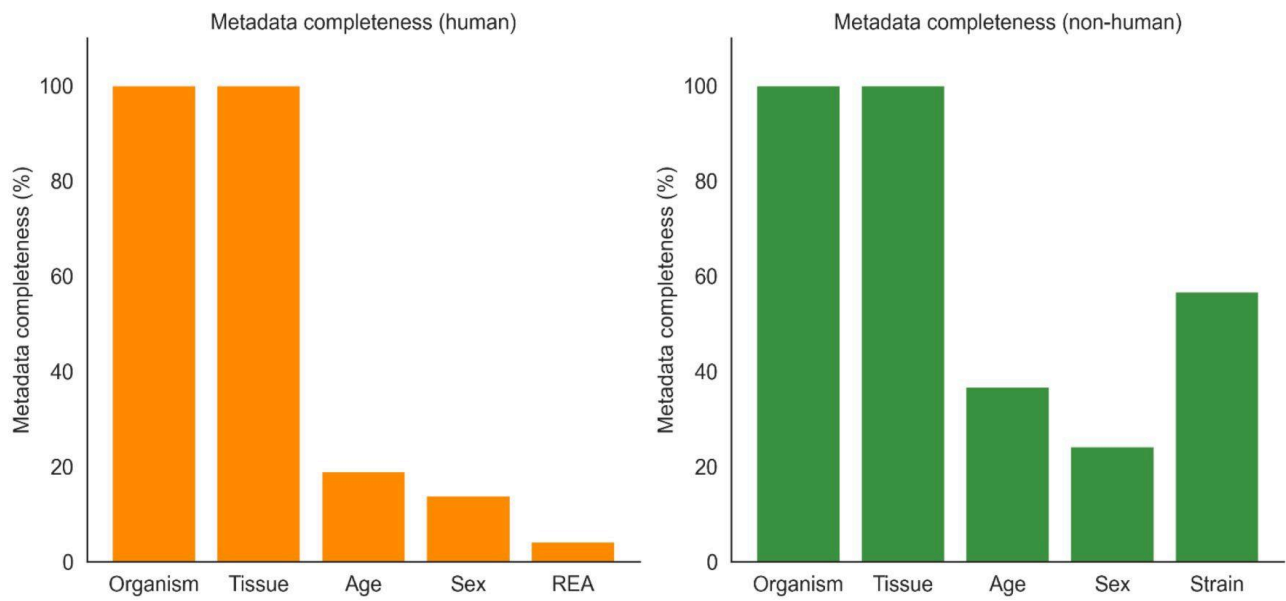


Figure S11: The proportion of organisms among the D2 dataset.

